To factorise an algebraic expression:

- Find the highest common factor and write it outside the brackets or outside the array.
- Divide each term by this factor and write the results inside the brackets or on top of the array.
- Check your result by expanding the brackets.

Here are some expressions that have been factorised:

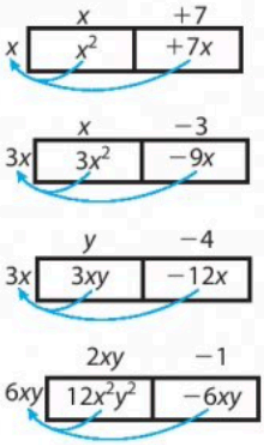
(i) $x^2 + 7x = x(x + 7)$

(ii) $3x^2 - 9x = 3x(x - 3)$

(iii) $3xy - 12x = 3x(y - 4)$

(iv) $12x^2y^2 - 6xy = 6xy(2xy - 1)$

Using the array method we have:



Exercise 4.1

1. Write down the highest common factor of each of these:

(i) 9 and 12 (ii) 12 and 18 (iii) 14 and 21 (iv) 21 and 35

2. Write down the highest common factor of each of these:

(i) $4x$ and $12x$ (ii) $3n$ and $9n$ (iii) $10x$ and $15x$

(iv) $3a^2$ and $6a$ (v) $3xy$ and $12x^2$ (vi) $2a^2b$ and $6ab$

Copy and complete numbers (**3–16**).

3. $4x + 8 = 4(\quad\quad)$

4. $6a + 12b = 6(\quad\quad)$

5. $7x + 14y = 7(\quad\quad)$

6. $22a + 33b = 11(\quad\quad)$

7. $12x - 24y = 12(\quad\quad)$

8. $ab + bc = b(\quad\quad)$

9. $2ax + 4ay = 2a(\quad\quad)$

10. $x^2 + x = x(\quad\quad)$

11. $5x^2 - 10x = 5x(\quad\quad)$

12. $3a^2 + 6a = 3a(\quad\quad)$

13. $4x^2 - 12x = 4x(\quad\quad)$

14. $6ab - 12bc = 6b(\quad\quad)$

15. $7x^2y - 14xy = 7xy(\quad\quad)$

16. $6x^2y - 15x^2 = 3x^2(\quad\quad)$

Factorise each of the following using brackets or the array model.

17. $4x + 16$

18. $6x + 18y$

19. $10a - 40b$

20. $ab + ay$

21. $3ab + 3bc$

22. $2xy - 2xz$

23. $7xy + 14xz$

24. $5ab - 15bc$

25. $6ax - 12ay$

26. $x^2 + 4x$

27. $3x^2 + 9x$

28. $5x^2 + 10x$

29. $6a^2 - 12a$

30. $7a^2 - 21a$

31. $10x^2 + 40x$

32. $25x^2 - 15x$

33. $7x^2 - 28x$

34. $12x^2 + 18x$

35. $15x^2 + 25xy$

36. $3x^2 - 6x^2y$

37. $x^3 + x^2 + x$

38. $3ab^2 - 6ab$

39. $12xy - 24x^2y$

40. $15x^3 - 35x^2$

Investigation:

Fully factorise each of the following expressions

(i)	$3a^2 - 15ab$	$2a^3 - a^2b$	$7ab - 35b^2$	
(ii)	$4a^2 - 2ab$	$2a^2b + 2a$	$2ab - 10b^2$	
(iii)	$7ab + 7b^2$	$5a - 25b$	$2ab^2 + 2b$	
(iv)	$4ab - 2b^2$	$3b^2a + 3b^3$	$a^3 - 5a^2b$	$2a^2b + 3ab^2$

Use the code below to find a letter for **each** factor.

Code

E	H	P	S	O	A	I	L	G	R	T	U	N
5	$2a$	$3a$	$2b$	$7b$	a^2	ab	$3b^2$	$a + b$	$a - 5b$	$2a - b$	$ab + 1$	$2a + 3b$

Rearrange each set of letters to spell a bird:

(i) _____ (ii) _____

(iii) _____ (iv) _____

Note: $3b^2a + 3b^3 = 3b^2(a + b)$